

CO1 AT1

Q36 Autonomous Vehicle Path Planning

A self driving car's path planning system use $T(n) = 3T(n/3) + n \log n$. Apply the master theorem to determine the time complexity. Carefully compare $P(n)$ and $n^{\log_b a}$ and identify the correct case and justify. Analyze how this recurrence affects for route optimization in dynamic environments.

Solution using Master theorem

Given

$$T(n) = 3T(n/3) + n \log n$$

compare it with the standard master theorem.

$$T(n) = aT(n/b) + f(n)$$

then

$$a = 3, b = 3$$

$$f(n) = n \log n$$

$$\text{Step-1: Find } n^{\log_b a}$$

$$f(n) = n \log n = O(n \log n)$$

since

$$f(n) = O(n^{\log_b a} \log^k n)$$

This matches master theorem case 2.

$$f(n) = O(n^{\log_b a} \log^k n), k=1$$

Step 3: Apply master theorem

For case 2

$$T(n) = O(n^{\log_b a} \log^{k+1} n)$$

Substitute the values

$$T(n) = O(n \log^2 n)$$

Final Answer

$$T(n) = O(n \log^2 n)$$

Interpretation

- The algorithm runs in $O(n \log^2 n)$ time
- It is more efficient than quadratic-time algorithms ($O(n^2)$)
- The extra $\log n$ factor comes from additional processing of each recursion level

Practical implementation for Autonomous vehicle route optimization

Efficiently computes optimal paths even for large road networks

Supports frequent route recalculations when traffic or road conditions change

The near-linear growth makes it suitable for real-time navigation system

Application: Insurance claim verification system
An insurance system uses three nested loops to verify policy details claim history and supporting records. Recall how triple nested loops affect complexity. Explain the rate of growth and determine both time complexity and space complexity for this process.

Ans ① Triple nested loops

A triple nested loop means one loop is inside another and third loop is inside the second. If each loop runs n times the total number of iterations is

② Rate of Growth

The running time increases cubically with the input size. As the number of policy details, claims records and supporting documents increase, the processing time.

③ Time complexity

The algorithm is

For ($i=0$; $i < n$; $i++$)

For ($j=0$; $j < n$; $j++$)

For ($k=0$; $k < n$; $k++$)

Each loop executes n times

Total Operations

$$n \times n \times n = n^3$$

Time complexity = $O(n^3)$

Space complexity

The algorithm uses only few loop variables (i, j, k) and no extra data structures

Space complexity = $O(1)$

⑤ Conclusion

Technique : Triple nested loops

Rate of growth : cubic

Time complexity : $O(n^3)$

Space complexity : $O(1)$

As the no of insurance policies, claims, histories and supporting records increases the verification time grow rapidly while memory usage remains constant